

Business Analytics Project Overview Report

Turning Your Vision Into A Defensible Plan A Technical Team Could Actually Build From

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BAN6800 – DATA ANALYTICS CAPSTONE | MODULE 2 ASSIGNMENT

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1. Define the Business Problem

The Module 1 Vision Document established that FirstBank Nigeria Limited, despite serving more than 42 million customer accounts, treats a highly diverse retail customer base with undifferentiated marketing and cross sell offers. This project overview applies the problem identification framework (define, gather data, analyze, root cause, objectives) to convert that vision into a plan a technical team could execute.

1.1 Define

The core business problem is low cross sell conversion and uneven progress on financial inclusion, caused by relationship managers and marketing teams relying on broad demographic proxies such as age and branch location instead of behavioral segments. This directly restates the problem framed in the Module 1 Vision Document and keeps this plan anchored to the same vertical, problem, and dataset.

1.2 Gather Data

As I documented in Module 1, the project uses the Kaggle Bank Customer Segmentation dataset (Bansal, 2021) as a structurally realistic proxy for FirstBank's core banking and card transaction data. The dataset provides over one million anonymized transaction records linked to roughly 800,000 customers, giving enough volume to profile behavior across demographic and transactional dimensions.

1.3 Analyze

A preliminary review of the dataset's transaction and balance fields shows customers with very different usage intensity and product needs are currently reachable only through the same generic campaigns, which is consistent with the marketing inefficiency described in the Module 1 Vision Statement.

1.4 Root Cause and Objectives

The section below applies a Fishbone analysis to trace this pattern to its underlying causes. The resulting objectives carry forward, unchanged, the measurable goals set in the Module 1 Vision Document: raise cross sell conversion by at least 15 percent, identify at least three underserved segments for financial inclusion, and maintain a quarterly fairness audit.

2. Root Cause Analysis

A Fishbone (Ishikawa) diagram was used to trace the low conversion and uneven inclusion problem defined in Section 1 back to contributing causes across four categories: data, process, technology, and people (Ishikawa, 1968).

Figure 2. Fishbone (Ishikawa) Root Cause Diagram

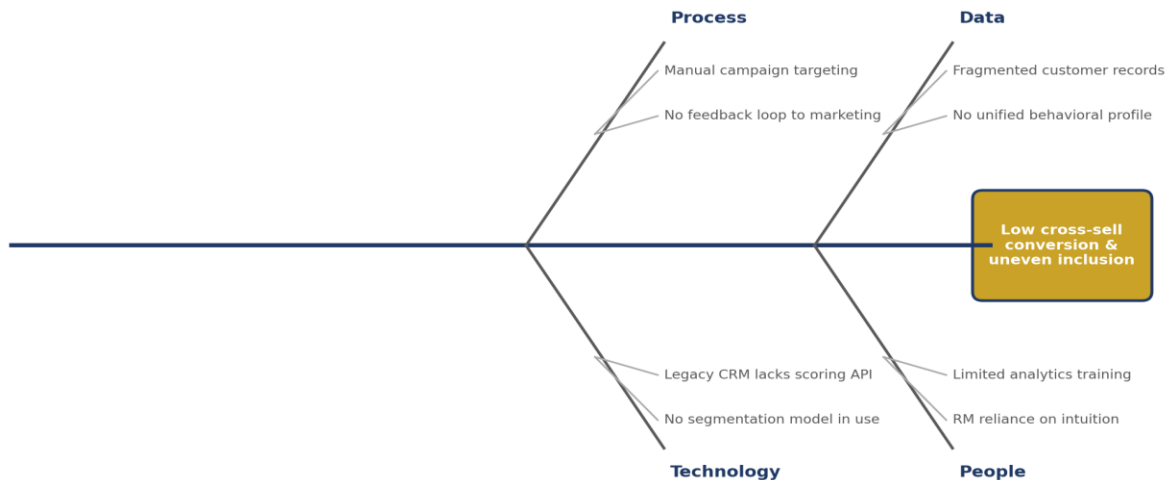


Figure 2. Fishbone (Ishikawa) root cause diagram for FirstBank's segmentation problem.

On the data side, customer records are fragmented across systems and there is no unified behavioral profile, which matches the dataset consolidation challenge acknowledged in the Module 1 Vision Document. On the process side, campaign targeting is manual and there is no feedback loop from campaign results back to marketing. On the technology side, no segmentation model is currently in production and the legacy CRM lacks a scoring API, which is precisely the gap the Module 1 technical stack was designed to close. On the people side, relationship managers rely on intuition rather than data, reflecting a limited analytics skill base.

2.1 Linking Root Causes to KPIs

- Fragmented data and no unified profile: measured by data completeness rate and time to produce a segment label, targeting the segment stability goal of 85 percent quarter over quarter set in Module 1.
- Manual, undifferentiated targeting: measured by cross sell conversion rate, targeting the 15 percent uplift goal from the Module 1 Vision Document.
- No segmentation model or scoring API: measured by model deployment status and API latency, both tracked in the RAID log in Section 5.
- Limited analytics skill and RM intuition reliance: measured by adoption rate of segment recommendations in branch conversations.

3. Business Analytics Project Overview

3.1 Objectives, Scope, and Deliverables

The objective is to deliver a working customer segmentation product, aligned to the vision, goals, and fairness objectives set out in Module 1, that a marketing and retail banking team can act on. In scope for this phase are data ingestion and cleaning, exploratory analysis, a baseline clustering model, a fairness and explainability layer, and a project board that sequences the remaining course modules. Out of scope for this phase are full production deployment and live integration with FirstBank's core banking systems, both of which depend on access not available in an academic setting. Deliverables include this report, an initialized GitHub repository following the Module 1 repository structure, an environment file, a Kanban project board, a technical RAID log, and a preliminary data dictionary.

3.2 Data Types and Sources

The project uses the same Kaggle proxy dataset identified in Module 1, combining demographic fields (age, gender, location) and transactional fields (balance, transaction date, amount, category). A preliminary schema is provided in the Data Dictionary in Section 7.

3.3 Analytical Methods

- Predictive: K-means and hierarchical clustering to assign behavioral segments, and gradient boosted classification to score product adoption propensity.
- Prescriptive: a rules based recommendation layer that converts segment and propensity scores into a ranked next best offer, as described in the Module 1 solution overview.
- Diagnostic: fairness auditing (Section 6) and SHAP based explainability to understand what drives each segment assignment.

3.4 Technical Architecture

The architecture below formalizes the technical stack proposed in Module 1, tracing data from ingestion through modeling to the dashboards and CRM systems that consume segment output.

Figure 1. FirstBank Customer Segmentation: Technical Architecture

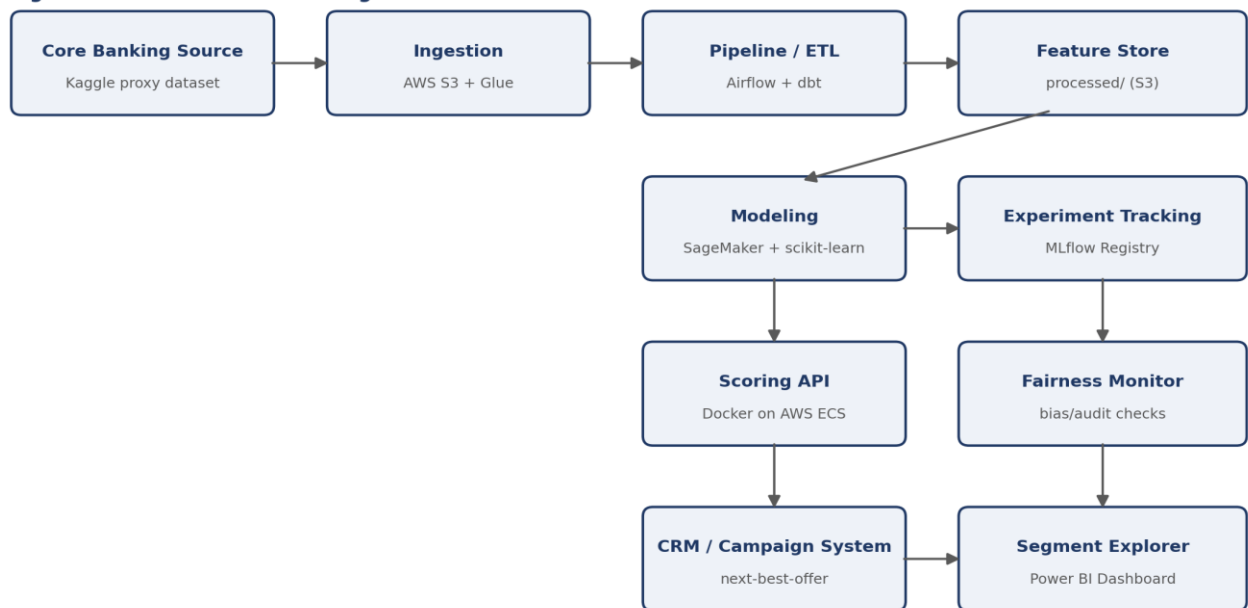


Figure 1. Technical architecture, consistent with the Module 1 technical stack matrix.

4. Applying Agile Methodology

The project is delivered in two week sprints following Scrum principles of iterative delivery, continuous stakeholder feedback, and adaptive planning (Schwaber & Sutherland, 2020). Each sprint ends with a review involving the stakeholders identified in the Module 1 stakeholder table, so that marketing and compliance feedback shapes the next sprint rather than being collected only at the end of the project. This iterative approach lets the team validate the segmentation logic on a small scale before it is proposed for wider use, which directly supports the fairness and accountability commitments in the Module 1 Ethical AI Vision.

4.1 GitHub Project Board

A Kanban board tracks work across Backlog, Sprint To Do, In Progress, Review, and Done columns. Tickets map directly to the folders proposed in the Module 1 repository structure, for example a ticket to build the ingestion DAG maps to src/pipeline, and a ticket to build the fairness audit script maps to src/fairness.

<https://github.com/OfiIreg/FirstBank-customer-segmentation>

Figure 3. GitHub Project Board (Kanban) Sprint 1 Snapshot



Figure 3. GitHub Project Board (Kanban) showing the Sprint 1 snapshot.

5. Implementation and Communication Plan

5.1 Stakeholder Roles and Communication

Stakeholder	Role (Module 1)	Cadence	Channel
Head of Retail Banking	Executive sponsor	Sprint review, biweekly	Steering committee meeting
Chief Marketing Officer	Business owner	Sprint review, biweekly	Demo plus written summary
Chief Data and Analytics Officer	Technical sponsor	Weekly standup digest	GitHub Project Board
Data Protection Officer	Compliance stakeholder	Per release	Ethics committee review

5.2 Monitoring Metrics

- Cross sell conversion rate, tracked weekly against the 15 percent uplift goal set in Module 1.

- Segment stability, tracked per refresh against the 85 percent quarter over quarter goal.
- Fairness audit pass rate, tracked quarterly against the Module 1 Fairness Objectives.
- Sprint velocity and ticket cycle time, tracked per sprint on the GitHub Project Board.

5.3 Technical RAID Log

Type	Item	Response / Owner
Risk	Proxy dataset schema may not fully match FirstBank's real core banking schema	Mitigate with a schema mapping document before Module 3; owner: Data Engineer
Risk	Cloud compute cost overrun on SageMaker during model tuning	Set budget alerts and use spot instances; owner: Chief Data and Analytics Officer
Assumption	The Kaggle proxy dataset is representative enough of FirstBank's retail base to validate the segmentation approach	Revisit once real data access is available; owner: Project Lead
Issue	Legacy CRM (identified in Section 2) has no scoring API yet	Build a lightweight API wrapper as an interim ticket; owner: Backend Engineer
Dependency	Fairness audit in Section 6 depends on protected attribute fields being available and correctly labeled	Confirm field availability with Data Protection Officer before Sprint 2; owner: Data Scientist

6. Privacy, Fairness and Compliance Requirements

This section builds directly on the Ethical AI Charter and Fairness Objectives set out in the Module 1 Vision Document, translating those principles into concrete technical requirements.

6.1 Data Privacy Plan

Customer identifiers are pseudonymized before they reach any analytics environment, consistent with the data minimization principle in Module 1. Access to customer level records is restricted through role based access control, and all data is encrypted at rest and in transit. Raw extracts are retained only as long as needed for model refresh, per the retention rule set in Module 1's Data Governance Principles.

6.2 Fairness Metrics

- Demographic Parity: segment assignment rates are compared across gender and age band to check that no group is systematically concentrated in lower value segments (Hardt et al., 2016).
- Equalized Odds: true positive and false positive rates for propensity scoring are compared across protected groups to check that model errors are not distributed unevenly (Hardt et al., 2016).

6.3 Explainability Framework

SHAP (SHapley Additive exPlanations) values are used to explain both the overall drivers of each segment and the specific reasons a given customer was placed into a segment, supporting the transparency commitment in the Module 1 Ethical AI Vision (Lundberg & Lee, 2017). Partial dependence plots supplement SHAP output for business stakeholders who need a simpler visual explanation.

6.4 Regulatory Compliance Checklist

Requirement	Status / Action
Nigeria Data Protection Act	Processing basis documented and reviewed by the Data Protection Officer identified in Module 1 (National Information Technology Development Agency, 2023)
GDPR (for diaspora customers)	Right to explanation supported through the SHAP based explainability framework in Section 6.3 (European Parliament and Council, 2016)
Central Bank of Nigeria guidelines	Segment driven offers reviewed to confirm no breach of consumer protection or pricing rules
PCI DSS	Card transaction fields handled under existing FirstBank cardholder data controls, not duplicated into analytics storage

7. Preliminary Data Dictionary

Field	Type	Description	Source
customer_id	String	Unique pseudonymized customer identifier	Kaggle proxy dataset (Bansal, 2021)
age	Integer	Customer age in years	Kaggle proxy dataset
gender	Categorical	Self reported gender	Kaggle proxy dataset
location	String	Customer's registered location	Kaggle proxy dataset
account_balance	Float	Account balance at time of record	Kaggle proxy dataset
txn_date	Date	Date of transaction	Kaggle proxy dataset
txn_amount	Float	Transaction amount	Kaggle proxy dataset
txn_category	Categorical	Transaction category or merchant type	Kaggle proxy dataset
segment_label	Categorical	Behavioral segment assigned by the clustering model	Derived, Module 3 onward

References

1. Bansal, S. (2021). Bank customer segmentation (1M+ transactions) [Data set]. Kaggle. <https://www.kaggle.com/datasets/shivamb/bank-customer-segmentation>
2. European Parliament and Council. (2016). Regulation (EU) 2016/679, General Data Protection Regulation.

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3. Hardt, M., Price, E., & Srebro, N. (2016). Equality of opportunity in supervised learning. *Advances in Neural Information Processing Systems*, 29.
 4. Ishikawa, K. (1968). *Guide to quality control*. Asian Productivity Organization.
 5. Lundberg, S. M., & Lee, S. I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, 30.
 6. National Information Technology Development Agency. (2023). *Nigeria Data Protection Act, 2023*. Federal Republic of Nigeria.
 7. Schwaber, K., & Sutherland, J. (2020). *The Scrum guide: The definitive guide to Scrum*. Scrum.org.